kurukshetra 2027™
The Battle of Brains
An International Techno - Management Fest

K! HACKS 4.0

ONLINE SUBMISSION

Team Name : Overclocked Engineers
College Name : Madras Institute of Technology
Problem Track : FinTech

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PROBLEM STATEMENT

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"...why your AI operated tools in the bank did not deem it fit to alarm him, that this transaction is suspicious?"

Chief Justice Surya Kant, Supreme Court hearing on digital-arrest scams, 9 Feb 2026 [2]

28 lakh

frauds reported on NCRP in 2025, up from 2.6 lakh in 2021 [1]

₹22,931 crore

reported lost in 2025 [1]

98.5%

of that money went in payments above ₹10,000 [1]

Most of this money is not stolen by hacking. The customer is talked into paying it.

- RBI says most cases are now authorised push payments, made with the correct PIN from the customer's own phone. Account takeover is "negligible" [1].
- Asking the customer often fails because victims are coached. The Mulki couple said they were buying land, and the Agra victim said it was for his son's property [3][6].
- RBI's proposed 1-hour lag would cover every payment above ₹10,000, and RBI itself notes fraudsters may make victims whitelist the payee [1].

Three recent cases

Navi Mumbai, Nov 2024 [5]
The manager noticed that a payee called "Royal enterprises" did not fit the customer's "medical" reason. ₹12 lakh was saved, after ₹21.94 lakh had already gone.

Agra, Sep 2026 [3]
An 80-year-old closed his FDs to send ₹81 lakh. The bank reached his daughter-in-law, who knew about the fake-police calls. The transfer was stopped.

Hyderabad, Oct 2026 [4]
An 80-year-old sent over ₹1.5 crore in six transfers over five days. The report mentions no bank stepping in.

Both saves came from a person at the branch who noticed something. A customer paying from an app has nobody like that.

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SOLUTION ABSTRACT

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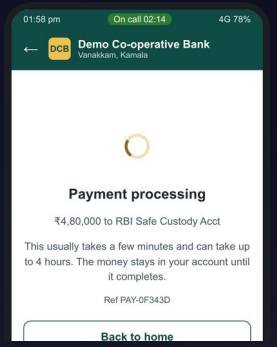
SafePause holds back only the risky payments, and the scammer on the call cannot tell.

It sits between the bank's app and its payment system and scores each outgoing payment in about 25 ms. It uses 26 signals the bank and phone already have, such as amount against the customer's usual, new payee, FD just closed, call from an unsaved number, pasted details and risk-listed receiver.

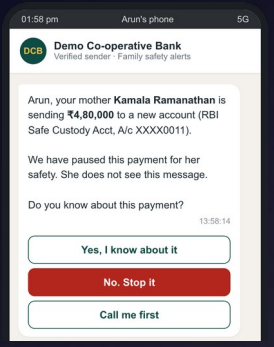
Low risk	Pays at once. This was 99.5% of genuine payments in our test.
Medium risk	Paused for up to 30 minutes while the app shows “processing”. The family member chosen at sign-up is asked if they know about the payment. Yes releases it, no sends it to an officer, and no reply releases it after 30 minutes.
High risk	Held for up to 4 hours. A bank officer sees the reasons in plain words, calls the customer, then stops it (money returned) or releases it.

RBI's options apply to everyone: a lag on every payment above ₹10,000, or a trusted person for every payment above ₹50,000 by anyone over 70 [1]. We use the same two tools only when the risk is high.

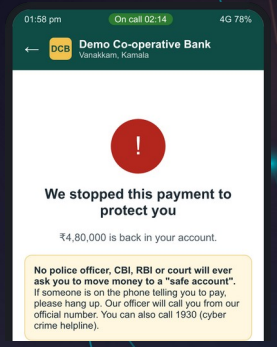
Only a person can stop a payment. RBI's FREE-AI report asks for human oversight of medium and high-risk AI, and says customers must be able to escalate to a human [13].



Kamala's screen



Message to her son



After the officer stops it

Result in our simulation (synthetic data)

	SafePause	1-hour lag
Scam money stopped	83%	11%
Genuine payments delayed	0.48%	4.6%

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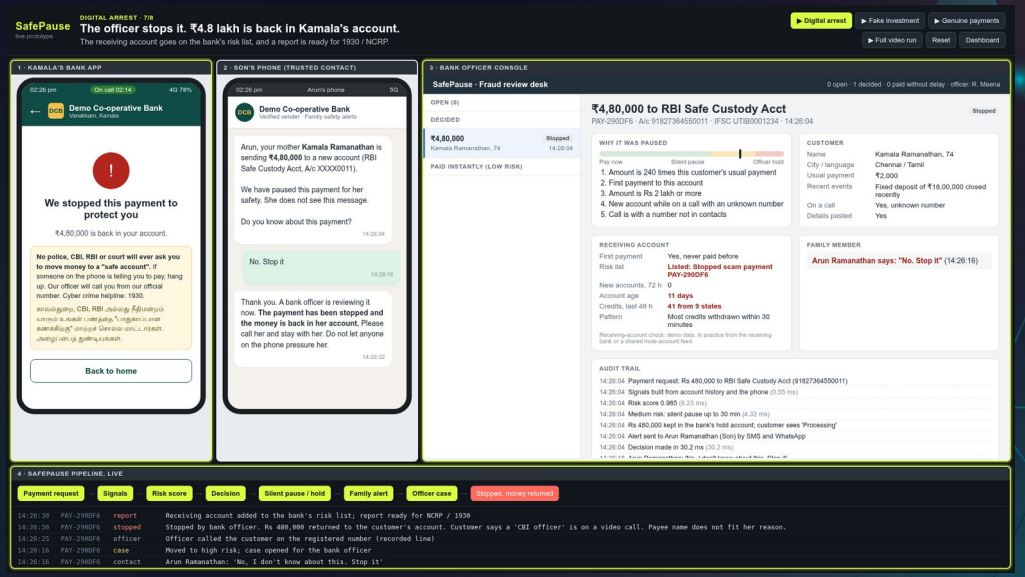
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ARCHITECTURE AND SYSTEM DESIGN

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Inputs	SafePause engine	Actions
Bank app SDK Call on? Number saved? Details pasted? Time to confirm, screen sharing.	**1. Signals** 26 signals from the inputs	**Hold** Money stays in the bank; app says “processing”
Core banking Payment history, known payees, FD closures, credits from strangers.	**2. Risk score** Scorecard model with reason codes	**Family alert** SMS or WhatsApp: Yes, No or Call me
Risk feeds DoT FRI risky numbers, mule-account lists.	**3. Decision** Low, medium or high, plus safety rules	**Officer console** Reasons, payee checks, call script

Afterwards the money is released or returned, the audit log is written, a report goes to NCRP / 1930, and confirmed cases are used to retrain the score.



Full code on GitHub: **github.com/Sanmugavel1/SafePause**

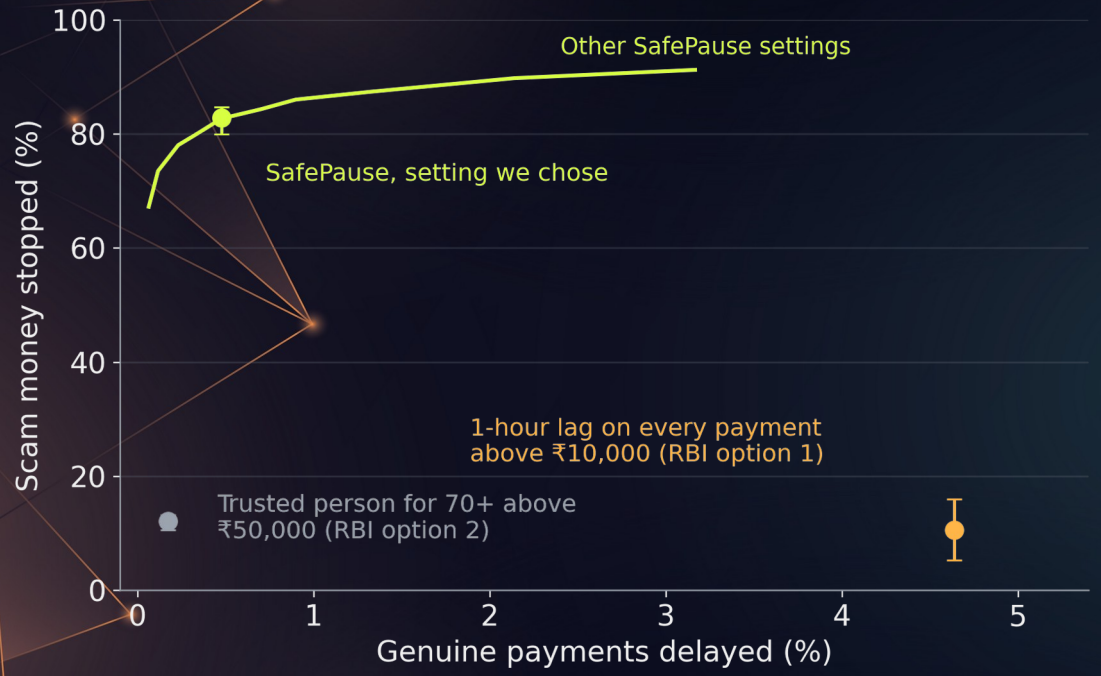
- **We built the complete working engine, not a mock-up:** app, live scoring, family alert, officer console and audit trail, on Python, SQLite and a REST API.
- Each decision took 17 to 30 ms on a laptop, using the same model as slide 5.
- Bank and phone data in the demo are simulated. No real bank is connected.
- No call audio is read. The phone checks the number against contacts and sends only “saved” or “not saved”.

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FEASIBILITY AND UNIQUE SELLING POINTS

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Synthetic test month: 2.79 lakh genuine payments, 2,731 scam cases, 200 runs over unknown human behaviour. Bars show the 90% range.

What the test showed

- Against a 1-hour lag on every payment, about 10 times fewer genuine payments were delayed and about 8 times more scam money was stopped.
- When the app openly said “held”, money stopped fell from 83% to 65%. We assumed the scammer then moves the victim to another bank.
- Task and job scams were left out of training, and 90% of that money was still stopped.

Published cases replayed (payment split assumed where not reported)

Case	Amount at stake	First paused at	If call not visible
Hyderabad Oct 2026	₹1.50 crore	payment 1 of 6	payment 2
Jodhpur Feb 2026	₹2.07 crore	payment 1 of 2	payment 1
Navi Mumbai Nov 2024	₹33.9 lakh	payment 1 of 4	payment 3

Similar steps already exist. UK banks may hold a suspicious payment for up to 4 business days since Oct 2024 [7], and Google warns Indian users who open finance apps while screen sharing on unknown calls [8].

Our weak spot is one-shot scams: only 39% (fake KYC call) and 35% (cloned-voice relative) stopped. All of this is synthetic, so a shadow pilot comes first.

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MARKET STUDY AND BUSINESS MODEL

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Who pays, and why now

- From 1 Jan 2027, a payment the customer makes under coercion counts as fraud, and the bank must prove negligence to refuse a refund [10]. Separate versions cover other bank types.
- The Supreme Court has asked for a shared-liability and compensation framework to be considered [2].
- A blanket lag costs goodwill. SafePause pauses about 1 in 200 genuine payments.

First customers

1,457 urban co-operative banks [11]

28 regional rural banks [12]

11 small finance banks [12]

They have older, rural customers and small fraud teams. UPI apps come next, then large banks through core-banking vendors.

DoT FRI scores phone numbers and MuleHunter.AI finds mule accounts on the receiving side. BioCatch, Feedzai and FICO sell platforms to large banks. We work on the sending side at the moment of payment, with a family member and an officer in the loop.

How we would charge

Our proposal, to be tested in the pilot

₹3 lakh one-time setup per bank

₹2 per mobile-banking user per month

Partners core-banking vendors bundle it; cyber-fraud insurers co-pay

Example: a bank with 1 lakh mobile users

Scam losses at national average	**about ₹4.1 crore a year**	₹413 per UPI user [1][9]
SafePause fee	**₹24 lakh a year**	about 6% of that
Officer time	**about 22 reviews a day**	one officer, from the simulation

Plan to get there

First, a 3-month shadow pilot with one co-operative bank, scoring only and pausing nobody. Then we compare the scores with its fraud reports, apply to the RBI Regulatory Sandbox, and move to paid roll-out.

RESEARCH AND REFERENCES

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How we tested it

- We built a synthetic bank from published cases: 30,000 customers, 60 days, 5.76 lakh payments and 6,000 scam cases of 6 types.
- Payments of ₹10,000 or more came out as 45% of scam payments by count and 98% by value, close to RBI's 45% and 98.5% [1].
- We trained on month 1 with 4 scam types and tested on month 2 with all 6. AUC was 0.98 on seen types and 0.97 on unseen ones. Hard genuine cases were kept in.
- We ran it 200 times with ranges for family replies, officer judgement and ignored warnings.

What others have said on record

"Banks must realise they are trustees of public money." CJI Surya Kant, 9 Feb 2026 [2]

Fraudsters had "digitally arrested the elderly couple for many days." Mangaluru Police Commissioner, Dec 2025 [6]

In-call warnings "helped thousands of users end calls that could have cost them a significant amount of money." Google, UK pilot, Dec 2025 [8]

References

[1] RBI, Discussion Paper: Exploring safeguards in digital payments to curb frauds, 9 Apr 2026
[2] Supreme Court: hearing of 9 Feb 2026 (LiveLaw); order of 4 Aug 2026, 2026 SCC OnLine SC 1532
[3] The420.in, Bank stops ₹81 lakh digital arrest transfer, Agra, Sep 2026
[4] The420.in, 80-year-old former banker, Hyderabad digital arrest fraud, Oct 2026
[5] Free Press Journal, Navi Mumbai bank manager saves 80-year-old, 29 Nov 2024
[6] ETV Bharat, Mangaluru manager saves couple from losing Rs 84 lakh, 5 Dec 2025
[7] UK Payment Services (Amendment) Regulations 2024; FCA guidance FG24/6
[8] Google Security blog, in-call scam protection for financial apps, 3 Dec 2025
[9] Lok Sabha reply: 55.49 crore UPI users, June 2026
[10] RBI (Commercial Banks - Responsible Business Conduct) Third Amendment Directions, 2026; ELP law note, June 2026
[11] RBI, co-operative banks publication, 29 Dec 2025
[12] Published lists of regional rural and small finance banks, mid-2025
[13] RBI, FREE-AI committee report, 13 Aug 2025
Replay also uses ETV Bharat, Jodhpur digital arrest case, 24 Feb 2026

Code, simulation and demo: **github.com/Sanmugavel1/SafePause**

Demo video: **https://youtu.be/fXHyqWcTbUs**

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TEAM DETAILS

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ROLE	NAME	STREAM / DEPT	YEAR	COLLEGE
Team Lead	**SANMUGAVEL B** Prototype, simulation, demo video	B.E. ECE Electronics Engineering	3rd Year	Madras Institute of Technology
Member	**MAATHESH THILAK K** Case research, interviews, market study	B.E. ECE Electronics Engineering	3rd Year	Madras Institute of Technology

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